

Assignment 3: AI System Development Proposal

A Multi-Modal Identity Verification System for Peruvian Elections

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Ethics 401, April 2026

The conversation around election security and validity has several facets, including issues of voter fraud and voter identity. With the advent of artificial intelligence in domains that include image capture and matching, voice identification, and more robust and accurate modeling, we propose development of an AI system that makes voting in Peru's national elections more reliable and more secure from a voter identity perspective. The proposed system is a multimodal AI-powered voter identity verification system using facial recognition, voice recognition, and fingerprint matching to authenticate voters at polling stations. As with any new AI system, the discussion must consider potential harms that the system might cause or exacerbate, risk mitigation strategies for said harms, and a recognition of unavoidable biases.

Potential Harms

Harms are assessed in several different ways. For the sake of brevity we will not anchor in any particular institutional prescription of harm assessment, and instead focus on several principles of ethical AI in a more generic format.

Privacy and Security Risks

More verification methods means more data collection. This could lead to different types of spoofing. For example, a person's voice can be recorded, relayed or synthesized using widely available tools (Nautsch et al., 2021). Similarly, fingerprint residue can be harvested and duplicated, and facial recognition is conceivably as simple as having a photograph of the spoofed person. Biometrics information could be used for nefarious purposes, such as creating deepfake videos or synthetic images (Tolosana et al., 2020).

From both a hardware and a software perspective, tools that support multiple authentication methods are more complex than those which perform only one task. A larger collection of hardware components (camera, microphone, fingerprint scanner) mean a greater risk of overall system failure (Microsoft, 2022). This is a particular risk in Peru's rural and remote polling stations, where technical support is likely to be limited and replacement hardware is not readily available (Carter Center, 2023).

The system will collect multiple types of data: voice, biometric, and fingerprint data. With more data comes increased security risk. The more data types the system collects, the greater the potential harm if the data is misused (Jain et al., 2016). A security breach exposing all three types of biometric data would be more problematic than if just one data type were breached. With all three datatypes compromised, additional layers of authentication can easily be defeated (Microsoft, 2022).

Reliability, Transparency, and Explainability Risks

Depending on which method(s) are required to fully authenticate a voter, there is a risk of verification failure. If all three methods are required to fully authenticate a voter and the voter fails one of the methods, does this mean the voter is now disqualified? For this reason, we are proposing that the system only requires one authentication method, with the other two being backups. Such a model reduces the risk of any person being turned away, but the risk is not fully mitigated. There is still the risk that all three methods will fail, in which case the system would direct the voter to speak to a human election worker.

Different verification models are sensitive to different environmental conditions. Peru's geography is diverse, meaning any polling station might encounter an environmentally related problem. Voice verification can degrade in noisy, crowded environments (Koenecke et al., 2020). Fingerprint

scanning can be affected by humidity or extreme dryness (Buolamwini & Gebru, 2018). Facial recognition may not recognize a tanned face or one covered in snow.

Bias and Fairness Risks

Peru's darker skinned Indigenous/mestizo majority is likely underrepresented in most AI model training data. Peru has millions of Quechua, Aymara, and Amazonian language speakers, all of whom are disadvantaged relative to native Spanish speakers (World Bank, 2015). Their exclusion might be treated as acceptable rather than a design failure that demands a non-biometric fallback option (Eubanks, 2018).

Risk Mitigation

While unavoidable circumstances will always remain and systemic biases cannot be eliminated entirely, developers can apply strategies early in the AI development life cycle to mitigate problems.

General Mitigation Strategies

These strategies involve general ML principles to ensure the system is fair, transparent, explainable, and accountable. Taking actions such as implementing locally representative training data and performing demographic bias audits before systems are deployed are now widely accepted as default standards to ensure the proper functioning of the system (NIST, 2019). There should also be a fallback to human review when biometrics fail. Finally, there must be ongoing monitoring since bias does not stay fixed. There needs to be a feedback and improvement cycle after each election to ensure continuous improvement.

System Design

For our specific use case, risks can be significantly mitigated via a biometric smart ID system where the voter's identity credentials and biometric templates are stored on a secure chip embedded in their ID card. At the polling station, live biometric data is captured and matched locally against the card, rather than against a centralized database.

On initial deployment, eligible voters would need to identify with the government through an existing process (license renewal, census, or similar) to verify their identity. During this process biometrics would be captured, with permission (Ammanath, 2022), including a fingerprint, voiceprint, and photo that are stored on a chip card similar to that now in use in Estonia (Republic of Estonia, 2022). The biometrics are not retained anywhere except on the chip card, to allow voters to retain sovereignty over their biometrics (Information Commissioner's Office, 2022; Ammanath, 2022). The person's identity would be stored as normal in the central registry (name, DOB, SIN, etc.), and their ID would also encapsulate this information.

When it comes time to vote, polls and poll workers are set up with biometric capture systems that read the chip card, and then match what's on the chip to samples provided, with permission, by voters. This live capture of biometric samples provided at the time of voting ensures that the person matches the ID and that the ID matches a central database. This reduces the risk of voter fraud, and reduces the chance of compromising voter identities. It would also reduce the chances of challenges to a voter's authenticity (for example challenging the legitimacy of a vote because, for example, signatures don't match between voter registration and ballot confirmation).

Having the biometrics stored on-chip means no external lookup is required. This is of particular advantage for remote regions where communication is sometimes more challenging or non-existent. Provided the equipment to capture biometrics for comparison, and the ability to read the chip, are functioning properly, voting can continue. The system also theoretically could be

deployed directly to voters by registered pollsters, perhaps saving those with mobility difficulties, or scheduling difficulties, from having to make a long trek. This of course requires additional consideration outside the scope of this paper, but it presents an interesting possibility for elections that rely on mail in ballots, or machine ballots.

By having the biometrics on the chip-card only, there is reduced risk of central surveillance, and no need to maintain a central biometrics database that would be very high risk (Republic of Estonia, 2022). The lack of a central database also means, in theory, limits on tracking when/where a person verifies their identity. No central database of biometrics also means a lower impact in the event of a breach (Ammanath, 2022; O'Neil, 2016). Biometrics cannot be readily ascertained if they are not present, even if a person has other PII, like DOB or SIN. The biometrics on the chip-card become another layer of security against identity theft or abuse of PII.

India's Aadhaar uses a similar system. Fingerprint and iris matching and a unique 12 digit ID number serve as a proof of residency (Unique Identification Authority of India (UIDAI),2023). Fingerprint matching is already conducted in Peru's elections (Registro Nacional de Identificación y Estado Civil (RENIEC), 2020), and would remain the primary method of identification, as it is a high-reliability assessment, backed by one or both of a voiceprint and facial recognition. By incorporating facial and voice recognition into Peru's voter identification procedure, the additional biometrics would help as multi-factor authenticator metrics, in the same way Aadhaar uses iris scans.

Having the different biometric options also serves as a backup in the event that any one option becomes unreliable or no longer viable (Information Commissioner's Office, 2022). Older Peruvian laborers tend to wear their fingerprints off over the course of their lives, for example, obviating the validity of fingerprints as viable metrics for identification (RENIEC, 2020). By including multi-modal authentication, it becomes much more difficult for an individual to pass as a deceased person with the argument that they've 'lost their fingerprint' (RENIEC, 2020).

System Resilience

The chip-card biometric identity verification process provides system resilience. This is critical in a nation like Peru where the rural regions do not have a massive redundancy in infrastructure (World Bank, 2021). Polling stations can operate offline, with the only information being stored being the ballots themselves, and a record of who has voted. This information is transported back to the capital after the election currently, so there isn't any change or disruption in the electoral process as far as that being a requirement. Being able to operate offline also means there's less dependence on vendor APIs and cloud deployments.

If the biometric matching is not sufficient to verify identity, or if consent was not granted to encode biometric data, or collect the data at the poll, a fallback protocol that requires human verification would still be available as an option

Realistically the system would perform a best 2 of 3 score matching, requiring a reasonably high degree of precision for the biometrics, especially given some of the inherent biases in facial recognition (National Institute of Standards and Technology, 2019). Using NIST's recommendations, rigorous pre-deployment testing and validation would need to be conducted to ensure latent bias does not unduly impact any facial recognition software, voiceprinting, or fingerprinting, especially as laborers, particularly in rural areas, will be exposed to higher degrees of sunlight and (at higher elevations) increased UV radiation, impacting melanin counts and causing skin to wrinkle, sag, and spot as citizens age. Manual labor and age degrade fingerprints, and smoke/dust, and other airborne particulate matter can cause vocal damage or disruption over time as well.

Operation and Maintenance

Given that people's physiology changes as they age, a regular update process to biometrics would need to be included as part of regular identity updates. At the very least, voluntary

updates/renewals would need to be available to account for changes in circumstances, such as oral or vocal damage, physical damage such as scarring, and loss of fingerprints. By allowing people the ability to update their biometric chip-card with the support of an approved institution, like a DMV or equivalent, the frustrations surrounding the system can be mitigated, and trust maintained in the system and the process, even when personal circumstances might have otherwise compromised a voter's ability to vote.

Regular re-training of the models will also be critical to ensuring the continued resilience of the system. As populations grow, and population patterns shift, it is important that the models being used grow and adapt to accommodate such changes.

Scheduled and random testing and audits of the system, including the models and of the biometric sampling units, will also be necessary to ensure continued robustness and to maintain trust.

Unavoidable Biases

When evaluating all the different voter identification methods, it's important to recognize that some bias is structural, and unavoidable, not just due to negligence. Distinguishing unavoidable bias from preventable bias is critical. If a bias is unavoidable, the goal becomes ensuring mitigation wherever possible.

Biometric Biases

As has been touched on above, each of the three voter identification methods under consideration (fingerprint, facial, or voiceprint recognition) face their own set of unique challenges. While each method generally works well enough for much of the population, there are some notable exceptions. Facial recognition underperforms on women and on darker skin tones (Buolamwini & Gebru, 2018; Grother et al., 2019). Fingerprint scanners struggle with worn, scarred, or aged fingertips. Voiceprint models underperform on community dialects and minority languages. Also,

ambient noise at voter stations can distort voice recognition, introducing an unavoidable issue. These issues and biases can lead to false rejection, further marginalizing groups that are already under-represented in the voter pool.

The obvious solution is to have an appropriately representative data set to train the models. The challenge comes from getting a large enough sample of underrepresented citizens to properly train such a model. Training robust biometric models requires large, representative datasets, making data collection and model development resource-intensive (Ammanath, 2022; Goodfellow et al., 2016). Depending on the level of dedication and resource allocation, this requirement may push into becoming untenable if the resource requirements become more intensive than the system's developers and supporters can reasonably bear, thus making the issue an unavoidable one.

Structural & Historical Bias

AI systems aside, there are distinct structural and historical biases to consider. Of Peru's population of 34 million people, over 20% of the population live in rural areas, which are often remote and harder to access. Communities are often spread out, and transportation can be difficult. As a result, voters in rural Peru often have to travel longer distances to reach their designated voting stations. This naturally often leads to lower voter turnout in rural regions compared to urban centers. Furthermore, the current system disadvantages communities with poor connectivity, unreliable power, or limited device quality. Rural and Indigenous communities in Peru disproportionately lack consistent electricity and internet access. These are technical failures resulting in further voter suppression. These realities reinforce the ongoing under-representation bias that these groups face.

Should the government of Peru elect to address this issue directly, in a manner similar to how low voter turnout is already addressed for remote and rural populations in several jurisdictions,

including India, Sweden, Canada, and Australia (Election Commission of India, n.d.; Australian Electoral Commission, n.d.; Elections Canada, n.d.; Valmyndigheten, n.d.), the proposed voter ID support and verification system proposed for Peru, by virtue of not requiring a network connection, and potentially supplying its own power, could further enhance the security and validity of such processes as they pertain to elections. These examples illustrate that physically bringing voting infrastructure to voters is a viable, though resource-intensive, approach to mitigating geographic barriers. By providing cost-effective voter verification systems, such initiatives could be made more viable in the future. There are of course tradeoffs with such a setup, including a potential reduction in privacy, as well as security and accountability considerations regarding the chain of custody for ballots, and the safety of polling workers.

Conclusion

The proposed AI system highlights the complexity of applying biometric identity verification in a real-world electoral context. While multimodal authentication can improve reliability and reduce fraud, it also introduces risks related to spoofing, system reliability, and data security, particularly in environments with limited infrastructure. These risks can be meaningfully reduced through a decentralized, chip-based identity system that minimizes data exposure and enables offline operation. However, even with these mitigations, limitations remain. Biometric systems are inherently imperfect, and broader structural factors, such as geography, infrastructure, and access, will continue to shape voter participation. As a result, the effectiveness of such a system depends not only on technical design, but on how well it integrates with the realities of the environment in which it is deployed.

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AI Usage Statement

AI was used to support brainstorming, structural organization, formatting, and refinement of this paper. All ideas, analysis, and final content decisions were reviewed and verified by the authors. The authors take full responsibility for the accuracy, integrity, and originality of the work.